

GRADIENT

The Intern Evaluation & Verified Portfolio System

Bridging the gap between assigned tasks and verified talent through continuous, data-driven performance tracking.



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1,020+ REVIEWS



Trusted by 1,000+ interns and teams worldwide

Problem Statement

Modern teams face systemic chaos when evaluating temporary talent

Blind Execution



Managers lack clear visibility into daily project status, overdue tasks, and task progression.

Subjective Endpoints



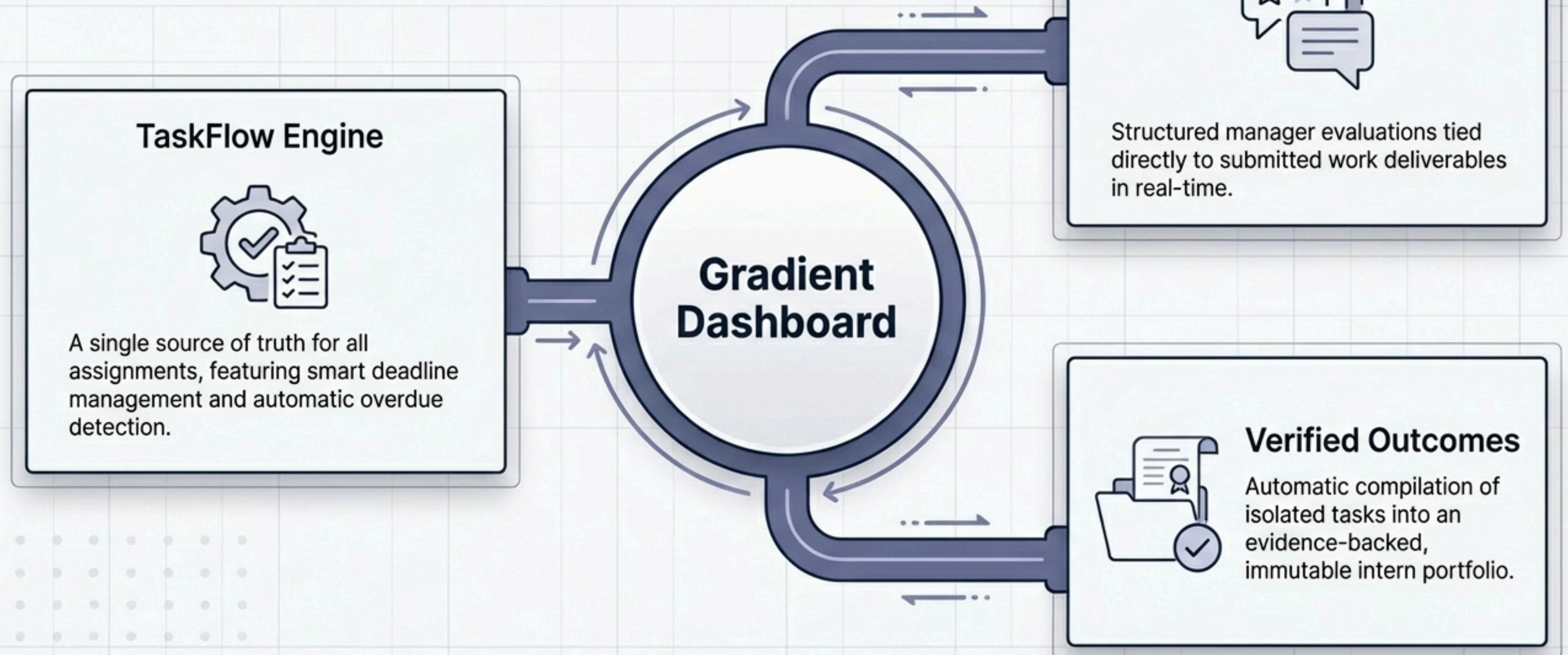
End-of-term reviews rely on memory and informal feedback rather than continuous, documented performance signals.

Scattered Assets



Work deliverables and critical documentation are lost across disjointed collaboration tools, local drives, and emails.

Gradient acts as a centralised engine for continuous evidence-based tracking



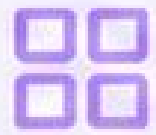
FR-01



User Authentication & Authorization

Secure user registration and login using email/password. Role-based access control for Admin, Manager, and Employee.

FR-02



Role-based Dashboards

Personalized dashboards for Admin, Manager, and Employee with role-specific data, analytics, and actions.

FR-03



Task Management

Create, assign, update, and track tasks. Set priorities, due dates, and statuses. Managers can monitor employee progress in real time.

FR-04



Real-time Chat & Collaboration

One-to-one and group messaging between users. Real-time communication using Socket.IO for instant updates.

FR-05



Portfolio Management

Employees can create and manage their portfolios, showcase projects, and update skills and experiences.

FR-06



Profile & Settings Management

Users can update their profile information, change password, and manage preferences. Profile changes are reflected across the system.

FR-07



Notifications & Alerts

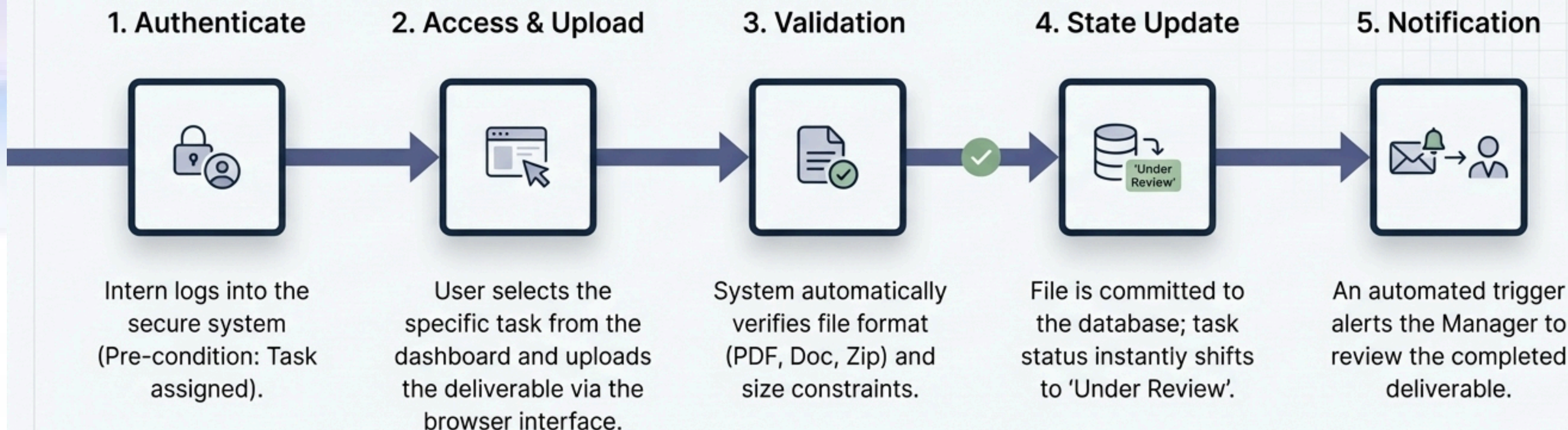
Real-time notifications for task updates, messages, mentions, and system alerts to keep users informed.

Strict role-based access control separates execution from administration

Capability	Employee/Intern	Manager	Administrator
Create/Manage Own Tasks	Full 	Full 	Full 
Provide Structured Feedback	Restricted 	Team Only 	Full 
View Analytics	Limited Personal 	Team Reports 	All Statistics 
User/System Management	Restricted 	Restricted 	Full Control 

Insight: This three-tier architecture ensures interns focus strictly on task execution whilst managers focus purely on evaluation.

The core user loop: Frictionless deliverable submission



A modern technology stack engineered for scale, speed, and safety

Client & State

React 18.3.1
(UI)

TypeScript 5.8.3
(Type-safety)

Vite 5.4.19
(Tooling)

TanStack Query 5.83.0
(Data fetching)

Styling & Analytics

Tailwind CSS
3.4.17

shadcn/ui & Radix UI
(Accessible components)

Recharts 2.15.4
(Visual analytics)

Backend as a Service

Supabase 2.86.0
(PostgreSQL)

Real-time
subscriptions

Secure file
storage

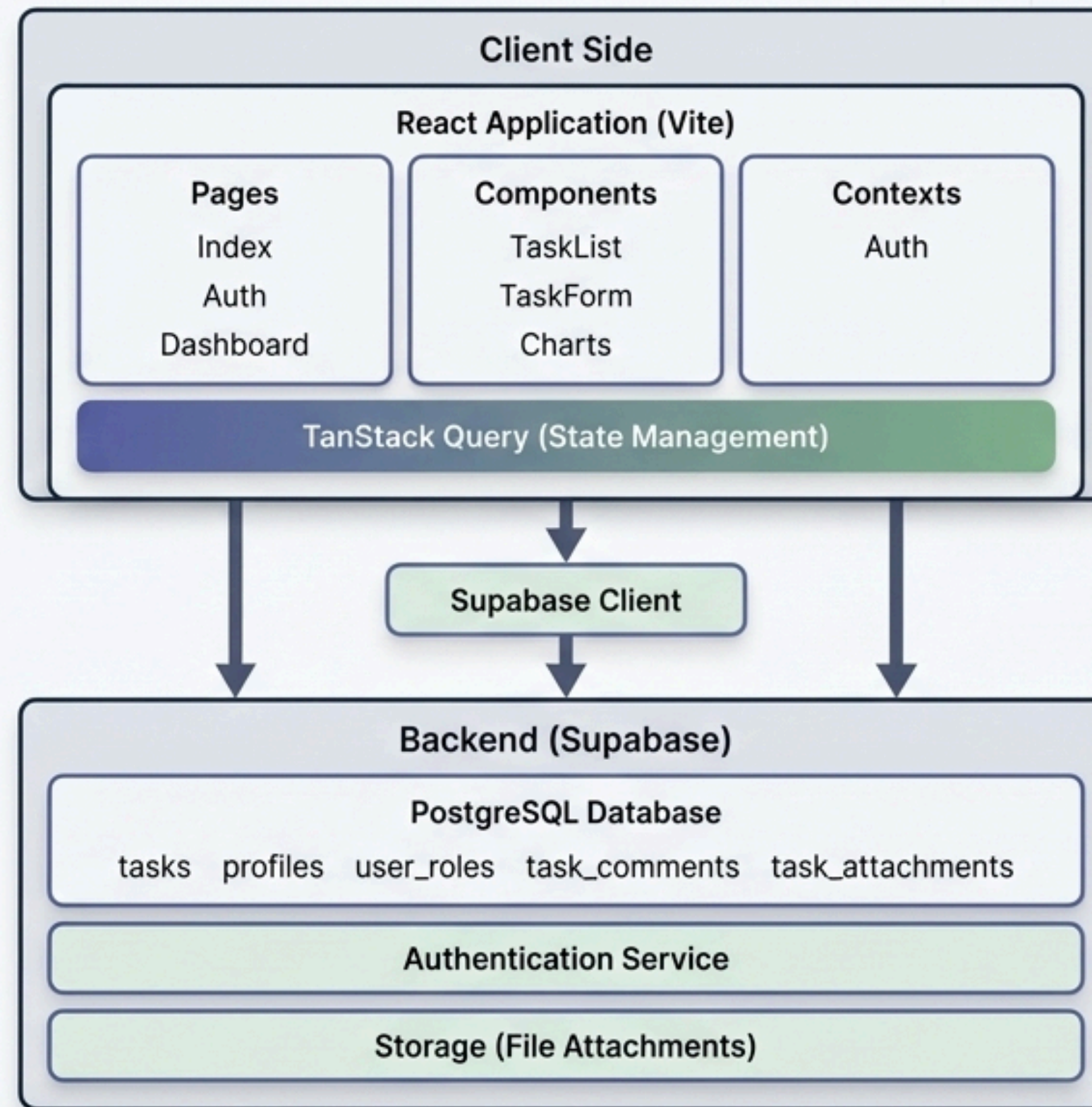
Row Level
Security (RLS)

The modular client-server architecture ensures separation of concerns

Top Layer (Client Side): Houses routing and isolated UI Components.

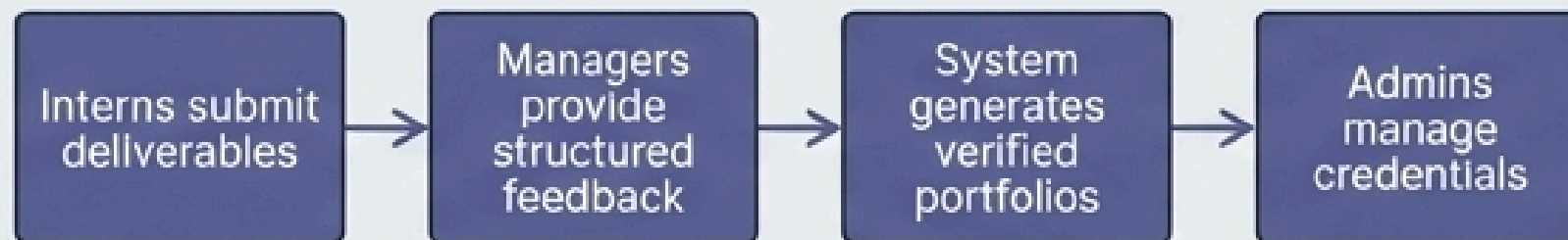
Middle Layer (The Bridge): Manages complex state and API requests.

Bottom Layer (Backend): Relational tables alongside dedicated Auth and Storage.



Software Requirements Specification establishes a resilient baseline

Q1: Functional Requirements



Q2: Non-Functional Guarantees

- JWT Authentication and HTTPS for security.
- API responses maintained at <500ms.
- Architected to support 500+ concurrent users.

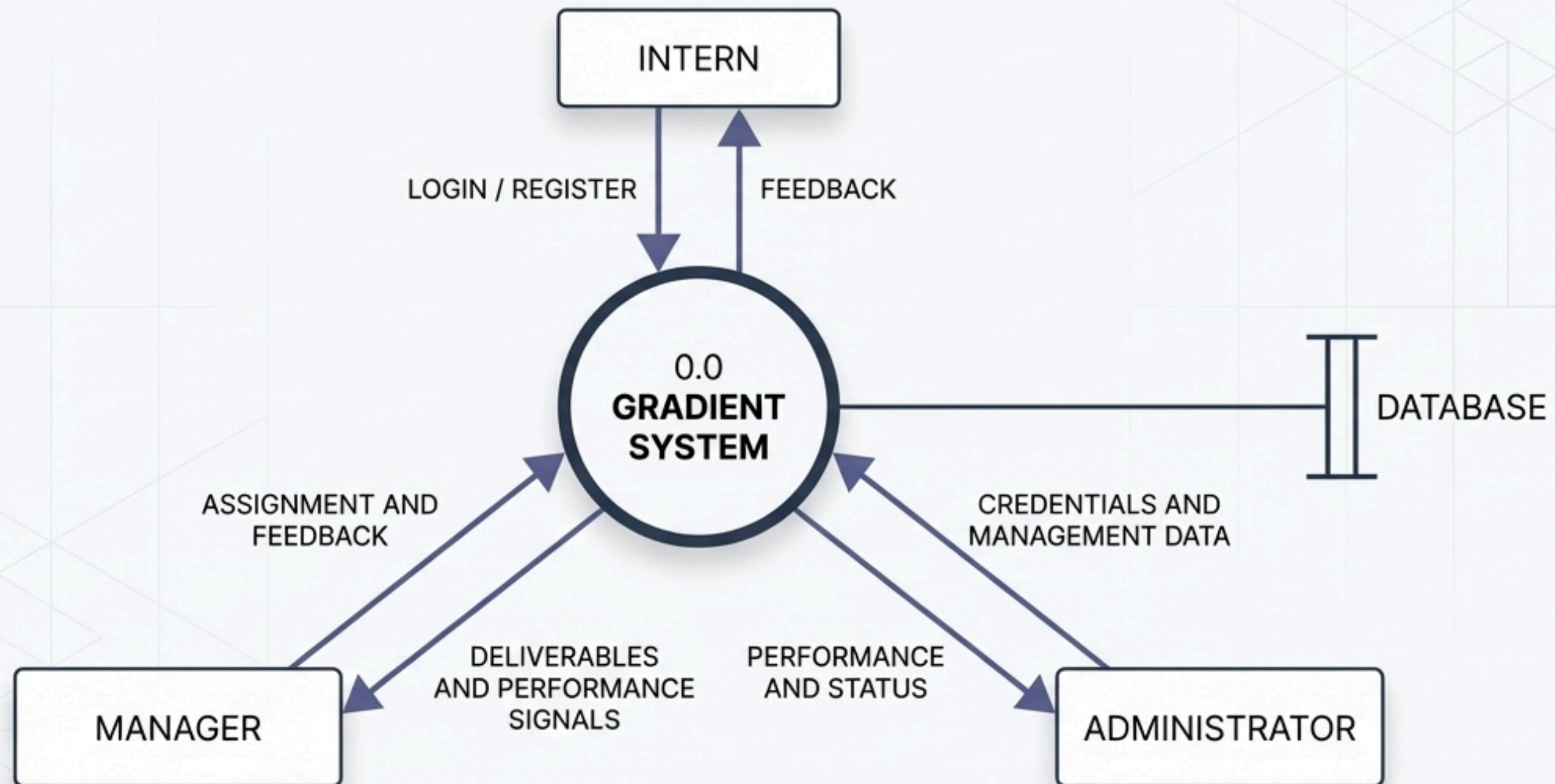
Q3: Operating Environment

- Client accessed via modern web browsers.
- Server runs on a Linux cloud environment with a Node.js runtime.

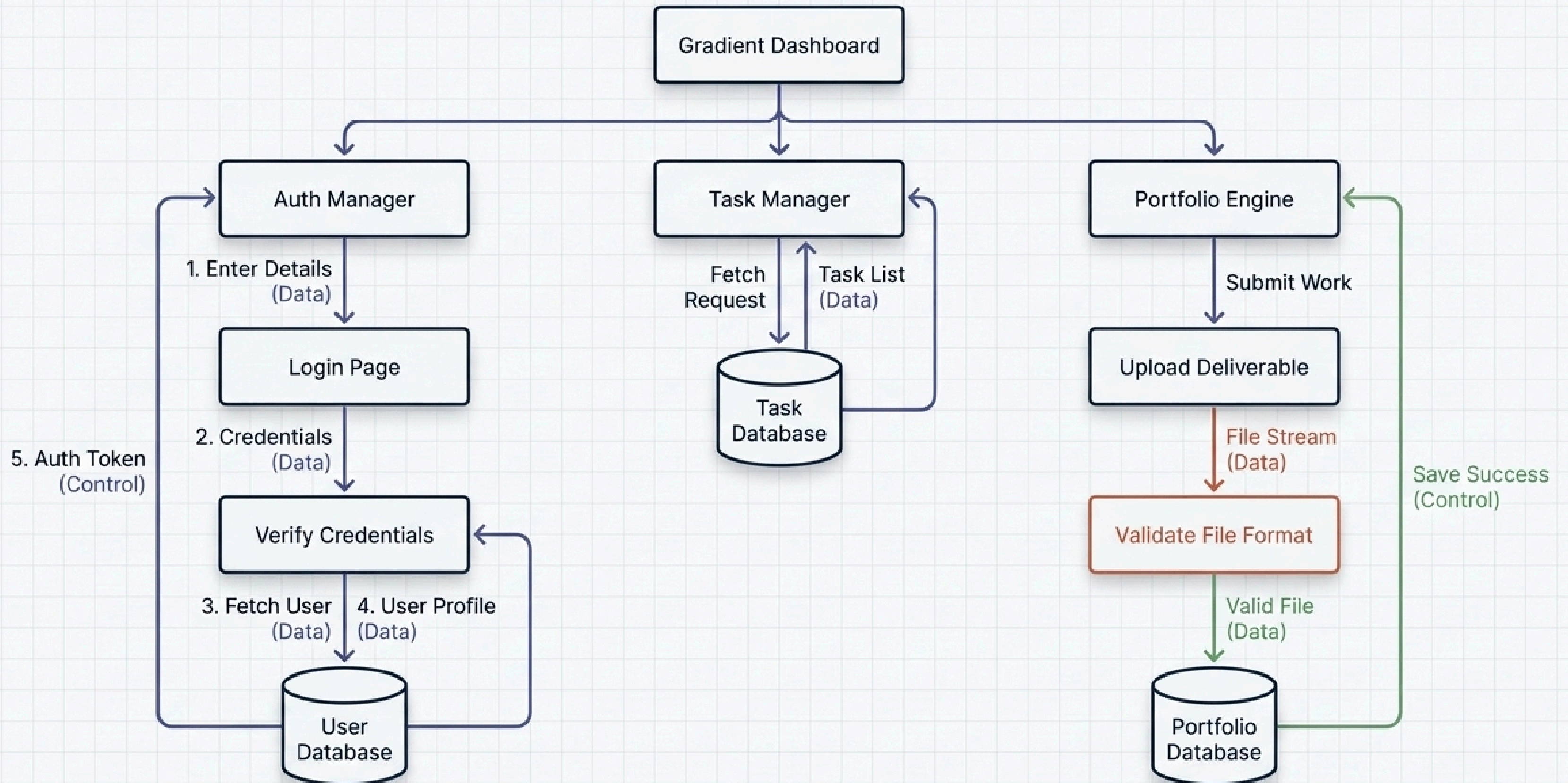
Q4: System Scope

- Operating purely as a transparent pipeline from initial task assignment to final talent identification.

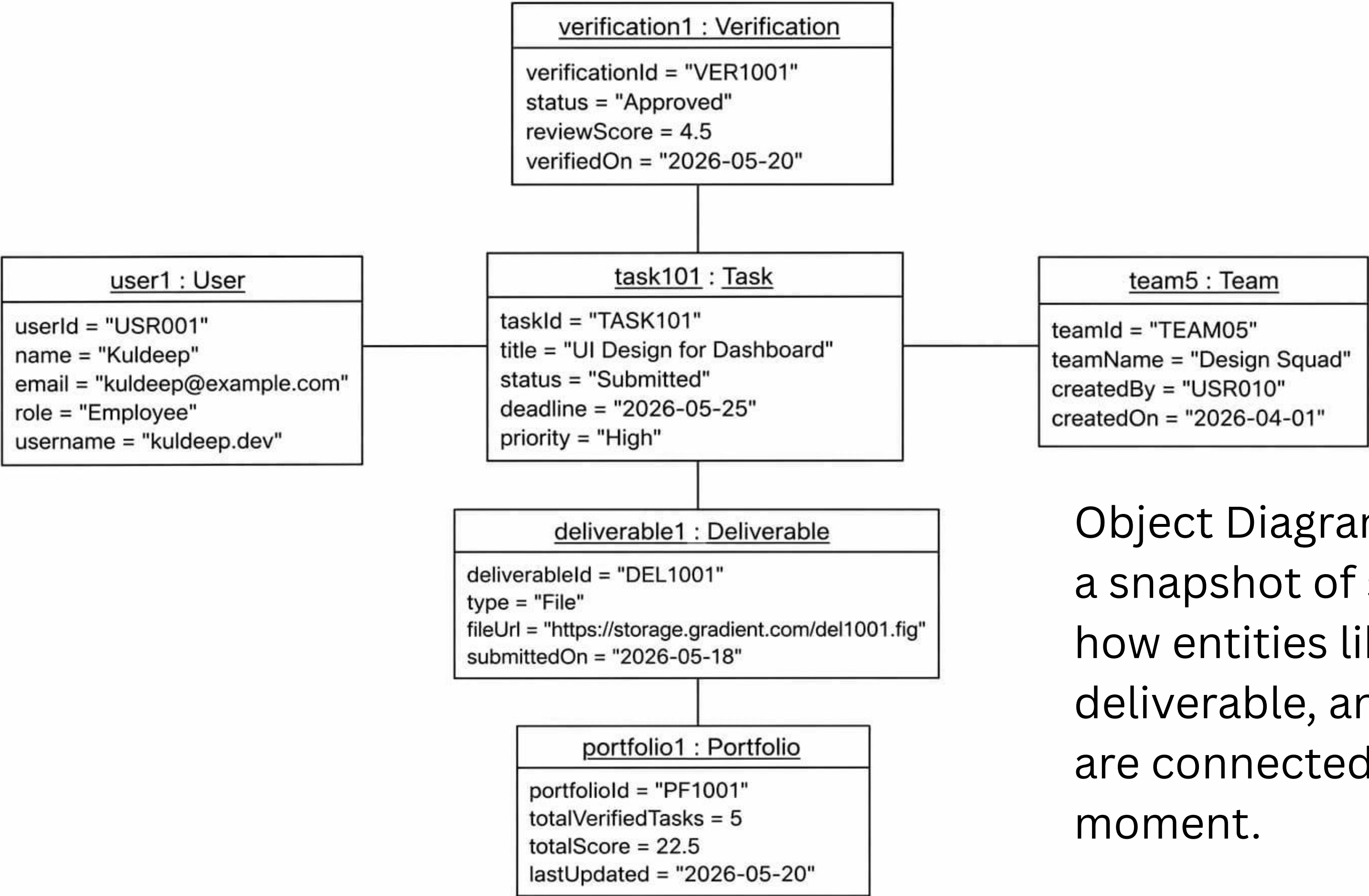
Macro Data Flow · maps the exchange between users and the central engine



Internal process routing separates authentication, tasks, and assets

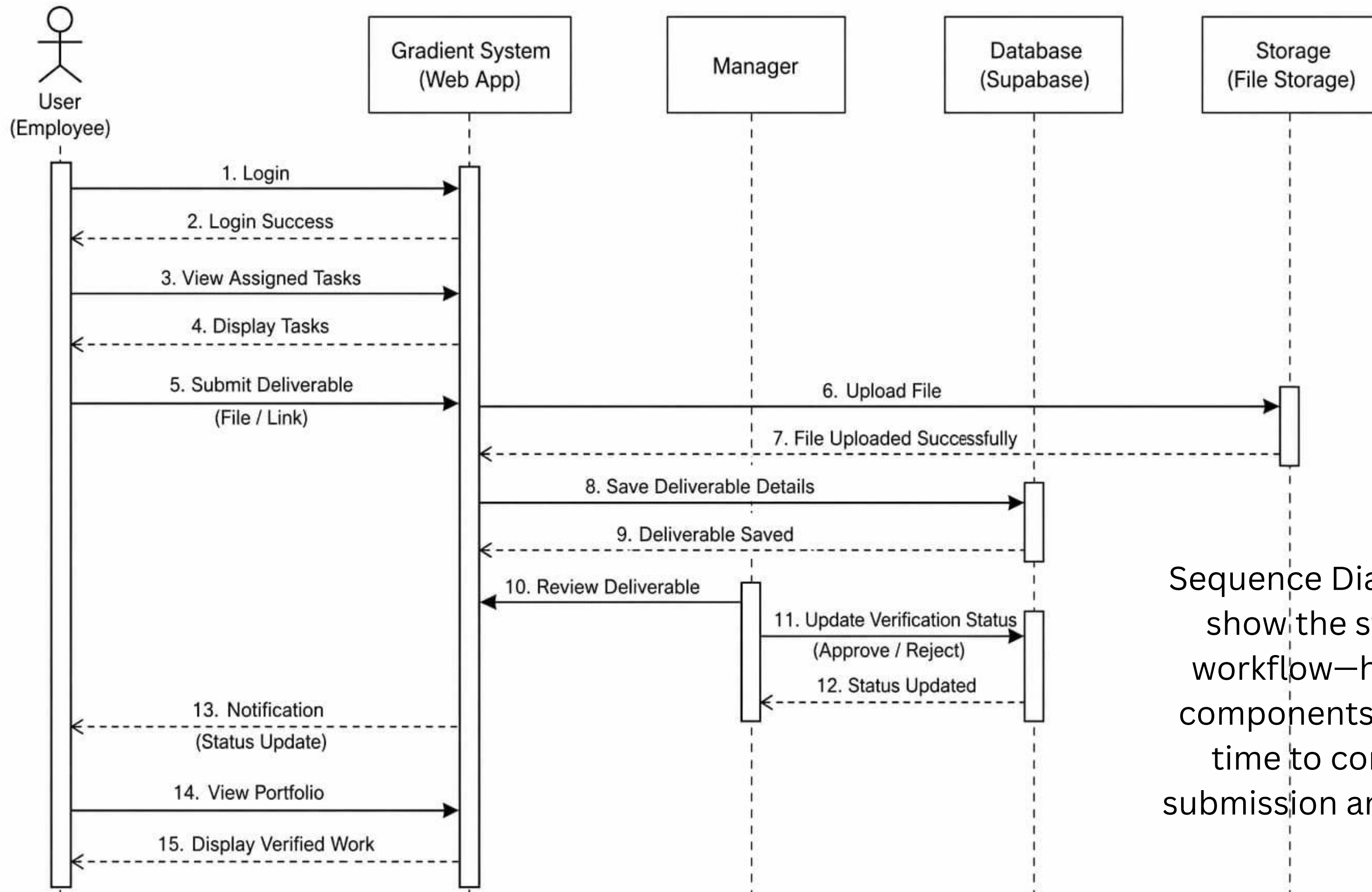


Gradient System: UML Object Diagram Snapshot

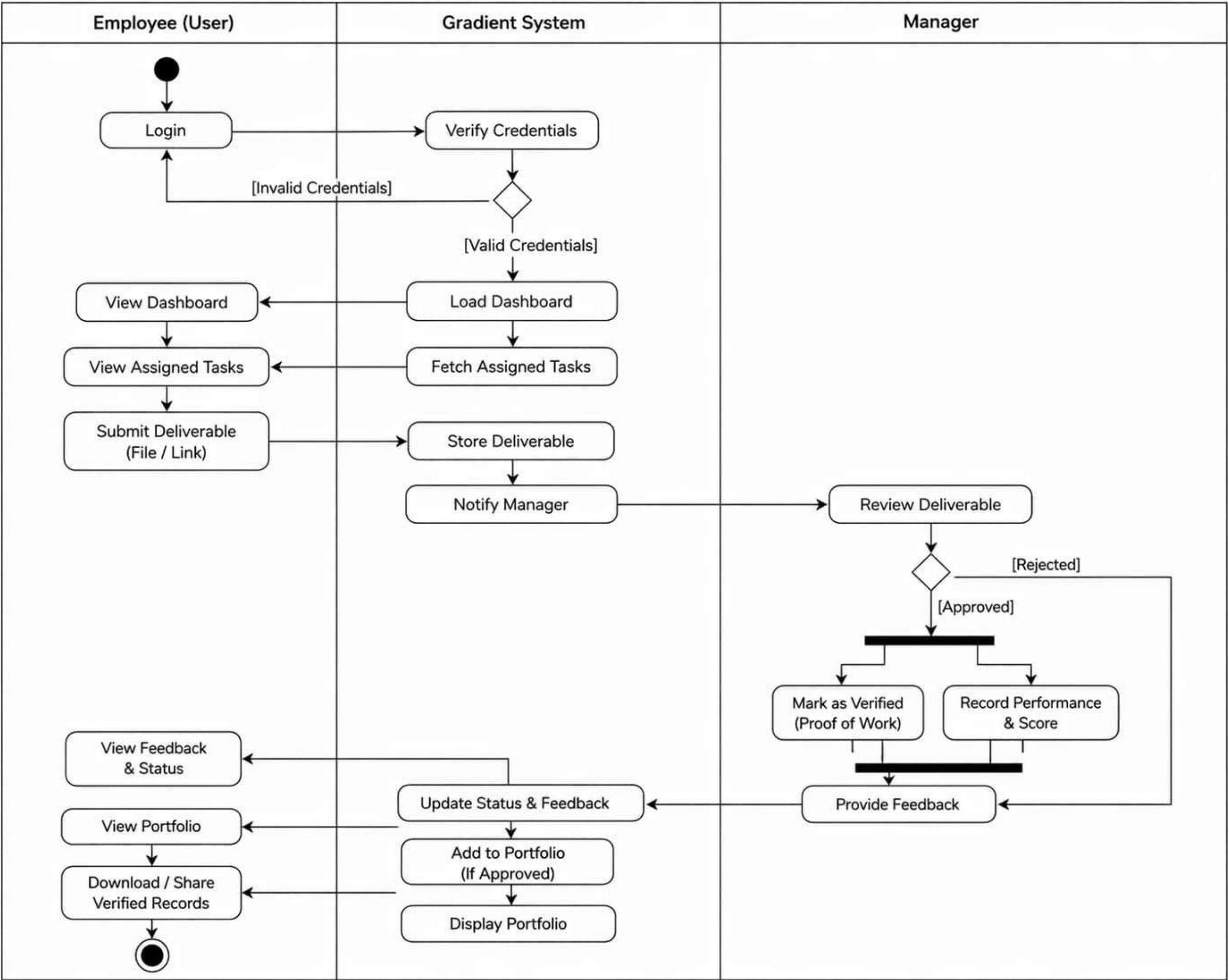


Object Diagram Goal: To show a snapshot of system data—how entities like user, task, deliverable, and verification are connected at a specific moment.

Sequence Diagram of Gradient Platform (Proof of Work System)



Sequence Diagram Goal: To show the step-by-step workflow—how different components interact over time to complete task submission and verification.



Activity Diagram Goal: To show the complete workflow of the system, including decision points like approval/rejection and how actions flow between user, system, and manager.

Database Schema

Relational model — PostgreSQL managed via *Prisma ORM*



Table: User

Field	Type	Description
id	UUID	Primary Key, Auto-increment (cuid)
name	VARCHAR	Full name
email	VARCHAR	Unique, Not Null
password	VARCHAR	Hashed password
role	ENUM	admin manager employee
avatarUrl	VARCHAR	Profile picture URL
bio	TEXT	Short bio about user
createdAt	TIMESTAMP	Auto-managed
updatedAt	TIMESTAMP	Auto-managed



Table: Project

Field	Type	Description
id	UUID	Primary Key, Auto-increment (cuid)
title	VARCHAR	Project title
description	TEXT	Project description
technologies	TEXT[]	List of technologies used
imageUrl	VARCHAR	Project cover image URL
githubUrl	VARCHAR	GitHub repository URL (optional)
liveUrl	VARCHAR	Live demo URL (optional)
createdById	UUID	Foreign Key → User(id)
createdAt	TIMESTAMP	Auto-managed
updatedAt	TIMESTAMP	Auto-managed



Table: Experience

Field	Type	Description
id	UUID	Primary Key, Auto-increment (cuid)
userId	UUID	Foreign Key → User(id)
company	VARCHAR	Company name
role	VARCHAR	Job role / position
startDate	DATE	Start date
endDate	DATE	End date (nullable)
currentlyWorking	BOOLEAN	Is currently working here
description	TEXT	Role description
createdAt	TIMESTAMP	Auto-managed
updatedAt	TIMESTAMP	Auto-managed



Table: Education



Table: Skill



Table: ContactMessage

Implementation Challenges & Resolutions

Key technical obstacles encountered during development and the solutions adopted

Challenge	Technical Solution	Outcome
 Frequent Logout & Session Inconsistency	Switched from localStorage to sessionStorage, centralised auth using AuthContext, and removed duplicate auth checks across components.	Stable user sessions across navigation and page refresh.
 Incorrect Role-Based Redirection	Fetches user role from Supabase 'profiles' table after login and implemented role-based routing (admin/manager/employee) using ProtectedRoute.	Users land on the correct dashboard based on their role.
 Real-time Notifications Not Received	Implemented Supabase Realtime (Postgres Changes) for tasks, chat, and payment updates with proper channel subscriptions.	Instant real-time updates for tasks, messages, and payments.
 File Upload Failures & Large File Handling	Used Supabase Storage with file size validation, compression, and unique naming to prevent overwrite and upload failures.	Reliable file uploads for deliverables, proofs, and media.
 Chat Messages Out of Order or Missing	Added created_at timestamps and ordered queries with <code>(.order('created_at', { ascending: true }))</code> for consistent message flow.	Chronological chat experience without missing messages.
 Task Assignment Not Syncing Across Tabs	Added Realtime listeners for task_assignments and profiles changes to sync updates across all open tabs.	Task updates reflect instantly across all user tabs.
 Inconsistent API Responses from External Services	Standardised API responses using mapping layer and error handling wrapper for all external API calls.	Consistent data structure and fewer frontend errors.
 CORS & Environment Issues in Production	Configured Supabase Edge Functions with correct CORS headers and environment variables for production.	APIs work seamlessly in both development and production.

Future Development Roadmap

Planned enhancements across three development phases



Phase I — Completed



User authentication & profile management



Role-based dashboards (Admin, Manager, Employee)



Task management & assignment



Real-time chat & collaboration (Socket.IO)



Portfolio management



Notifications & alerts



Reports & analytics (basic)



Profile & settings management



Phase II — Near-Term



File uploads & document management



Calendar integration & deadline reminders



Advanced reports & export options



Team management & performance tracking



Project templates & reusable workflows



Cloud storage integration (Google Drive / Dropbox)



Mobile responsive improvements & PWA support



Two-factor authentication (2FA) for enhanced security



Phase III — Long-Term



AI-powered task suggestions and smart prioritization



AI chat assistant for project and task queries



Third-party integrations (Slack, GitHub, Jira, etc.)



Advanced analytics & predictive insights



Multi-language & regional support



Customizable dashboard widgets



Public API & developer extensions



Enterprise features: SSO, Audit logs, Advanced roles

Conclusion

The project successfully demonstrates a complete proof-of-work system that ensures transparency, accountability, and structured task tracking.

It integrates user actions, system processing, and managerial verification into a seamless workflow from submission to approval.

Overall, the platform enhances productivity and builds a reliable portfolio system by validating and showcasing verified work.